Féidearthachtaí as Cuimse
Infinite Possibilities

AI: Theory to Practice

Data Understanding and Preparation

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Plagiarism declaration:

We confirm that this report is our own work. All external sources, tools, and assistance (including GenAI) have been appropriately acknowledged.

Contribution statement:

Max Sri-Uttha X00213723 – 33 % : Identify dataset, Define problem, Define target and features, Real-world relevance, Data shape & types, Summary of features

Odhran Chamberlain X00217322 – 33 % : Visual analysis, Insights from visualizations, General data quality check, Handle missing values, Handle outliers

Oleh Kravets X00219607 – 33 % : Project Management, Coding, Documentation

Dataset source: https://huggingface.co/datasets/mstz/nbfi

Dataset license: CC0: Public Domain

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Data Understanding (EDA)

In this section we analyze the raw dataset (**df_raw**) before any cleaning. We focus on descriptive statistics, distributions, missingness, outliers, and relationships between variables. We also flag data quality issues that will need to be fixed in Data Preparation. During inspection we also noticed some columns that were technically numeric but are really codes or categories (e.g. weekday codes). We handle that later.

Statistical summaries

We generated descriptive stats for all numeric columns in df_raw. For each numeric column we calculated: **count, missing, unique, means, medians, modes, cardinalities, standard deviation, percentiles, min, max, range, IQR, etc.**

For each categorical column we report: **number of unique categories, missing count and % missing, most common category (mode), how often that mode appears, what % of the rows that is.**

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Data Understanding (EDA)

Numeric feature summary

	count	mean	std	min	25%	50%	75%	max	iqr	n_unique	missing	missing_%
income	8308.0	19423.244179	11384.515832	3600.000000	12600.000000	16812.000000	22500.000000	225000.000000	9900.000000	232	0	0.0
nr_children	8308.0	0.549711	0.779266	0.000000	0.000000	0.000000	1.000000	12.000000	1.000000	9	0	0.0
credit	8308.0	64350.704853	42933.182637	4500.000000	28898.887500	54000.000000	88900.875000	315000.000000	60001.987500	1625	0	0.0
loan_annuity	8308.0	2929.035430	1542.174857	297.900000	1809.337500	2670.300000	3751.650000	22500.000000	1942.312500	3636	0	0.0
education_level	8308.0	1.405633	0.509098	0.000000	1.000000	1.000000	2.000000	4.000000	1.000000	4	0	0.0

- income is strongly **right-skewed**. Median is reasonable, max is extremely high (200K+). The mean > median, so the tail is pulling the average up.
- credit (requested loan amount) ranges from small to huge (300K+). This matters a lot commercially because the lender's exposure lives here.
- loan_annuity (repayment per period) behaves the same way: it has a long right tail.
- consecutive_days_of_employment shows that some people have extremely long uninterrupted employment histories. Statistically they get flagged as **"outliers"**, but that's stable employment, not necessarily dirty data.
- score_2 is interesting. For most rows it looks like a 0–1 scale, almost like a probability. But there is a row with score_2 = 100. That's clearly **off-scale**.

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Data Understanding (EDA)

Categorical feature summary

	count	n_unique	mode	mode_freq	missing	missing_%	mode_freq_%
owns_a_car	8308	2	False	4975	0	0.000000	59.882041
owns_a_bike	8308	2	False	5427	0	0.000000	65.322581
has_an_active_loan	8308	2	True	4247	0	0.000000	51.119403
owns_a_house	8308	2	True	5672	0	0.000000	68.271545
accompanied_by	8308	6	Alone	6888	0	0.000000	82.90804

Some features are **constant** in our dataset. For example:

- is_male is always False for every row we have.
- has_provided_a_mobile_number is always True.
- same_residence_and_home, same_work_and_home, and nr_defaults_in_social_circle also do not vary.

Constant columns have zero variance - they are not useful to the model and will be dropped.

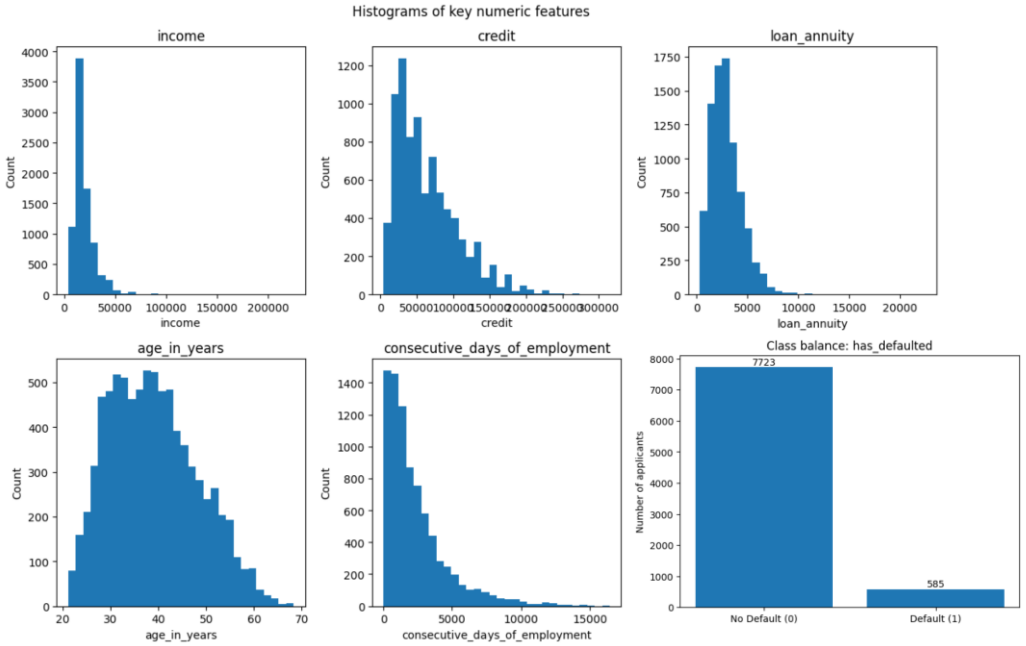
- The target has_defaulted is highly **imbalanced**. Only around 7% of rows are 1 (defaulted). The rest are 0. That imbalance will matter later.

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Data Understanding (EDA)

Visual analysis

Histograms and bar plots

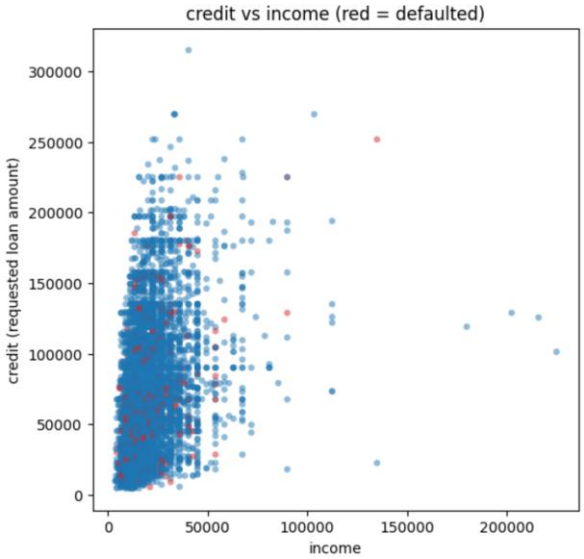


- the financial features (income, credit, loan_annuity) are all right-skewed with heavy high-end tails. A small number of applicants request or can afford very large values.
- consecutive_days_of_employment also stretches far to the right, which reflects long-term stable employment.
- only ~7% of rows are defaults. Any future model will have to handle class imbalance, because predicting “always 0” would look accurate but be meaningless.

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Data Understanding (EDA)

Pairwise scatter

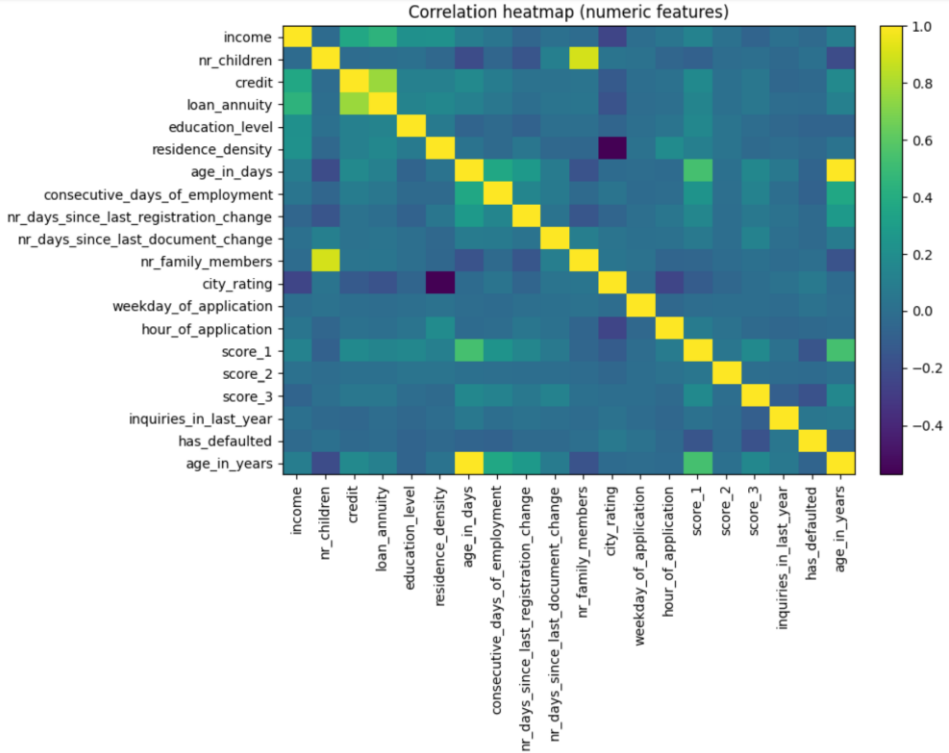


- Higher income tends to come with higher requested credit. That's expected.
- Red points (defaults) are not just clustered in the low-income / low-loan corner. They appear across the range. That tells us we might need affordability-type ratios (like credit / income or loan_annuity / income) instead of just raw values.

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Data Understanding (EDA)

Correlation heatmap

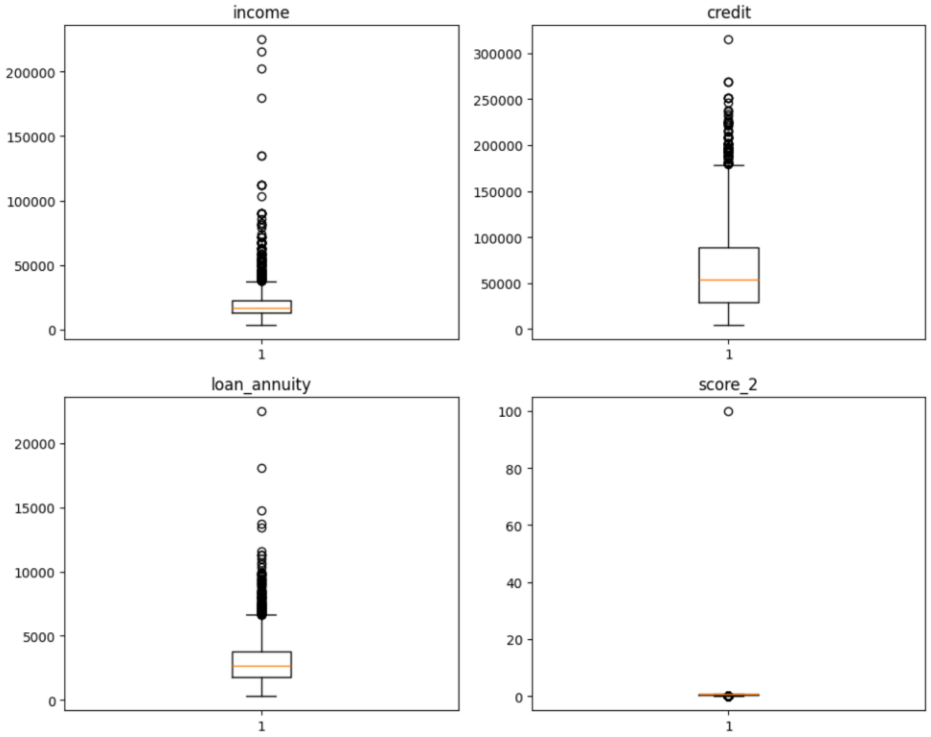


- nr_children and nr_family_members have extremely high correlation (about 0.9). They're basically the same household-size signal.
- credit and loan_annuity also correlate strongly (~0.76), which is natural: bigger loan → higher repayment.
- city_rating and residence_density have extremely low correlation (about 0.1). It is normal.

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Data Understanding (EDA)

Outliers and suspicious values

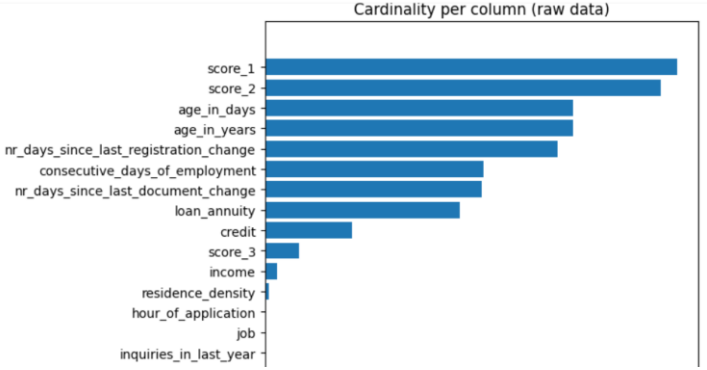


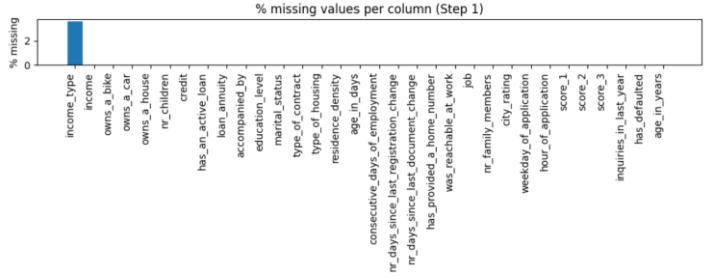
- we see extremely large values for income, credit, and loan_annuity. These are not necessarily “bad data.” They might just be high-exposure
- score_2 is weird: almost all rows are in [0,1], but one row has 100. That one row is definitely off-scale.

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Data Understanding (EDA)

Cardinality and missing values





- column with very high cardinality are not useful to the model and can be dropped.
- only income_type has missing values (~3.4%).
- Instead of dropping those rows, we keep them and fill income_type with “Unknown”.
- We want “Unknown” to survive as a separate category so the model can learn from it.

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Data Preparation

In this section we take the raw data and make it “model-ready.” We do this in explicit, documented steps. In the notebook, we save intermediate versions so that each step is traceable:

df_raw → original

df_step1 → after basic cleaning

df_step2 → after handling missing values

df_step3 → after handling outliers/extremes

df_step4 → after reducing redundancy

df_step5 → after encoding and feature engineering

X_ready, y → final ABT for modelling

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Data Preparation

General data quality check

Duplicates

We found 437 exact duplicate rows in df_raw. We removed them.

- Before removing duplicates: 8,308 rows
- After removing duplicates: 7,871 rows

This prevents the same customer/application from being counted multiple times.

Unimportant columns / constant columns

Some columns have the same value for everyone:

- is_male
- has_provided_a_mobile_number
- same_residence_and_home
- same_work_and_home
- nr_defaults_in_social_circle

A feature with zero variance can't help a model distinguish between default and no default. We drop them.

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Data Preparation

Incorrect data types

We fixed types for certain columns:

- hour_of_application: cast to int (0–23).
- education_level: keep as integer since it's ordered.
- city_rating: keep as integer since it's ordered.
- weekday_of_application: treat as category, not a numeric quantity.

After these changes we call the dataset df_step1.

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Data Preparation

Handle missing values

We quantified missingness. Only one feature mattered: income_type (about 3.4% missing).

We then checked the default rate in:

- rows where income_type was missing,
- rows where it was provided.

The missing income_type rows showed slightly higher default risk. That means “did not provide income type” is itself a meaningful behaviour.

Decision:

- We do not drop those rows.
- We fill missing income_type with the label "Unknown".
- We keep "Unknown" as its own category going forward so the model can learn that pattern.

After we perform that imputation, we call the dataset df_step2.

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Data Preparation

Handle outliers

We do not just delete “outliers.” Instead, we transform them so that:

- the model is not dominated by extreme numeric values,
- high-risk, high-exposure rows remain in the dataset.

We apply two main strategies:

- Cap + log transform on money features
- Fix broken scaling in other columns

After applying these transformations and creating these engineered columns we call the dataset df_step3.

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Data Preparation

Check for multicollinearity

Some pairs of features are almost the same information. The big one:

- nr_children vs nr_family_members have correlation around 0.9.

If we feed both into a model that tries to assign individual importance to each column, that high redundancy can cause unstable behaviour. We don't need two columns saying the same thing.

Action:

- Drop nr_children.
- Keep nr_family_members.

We also saw credit and loan_annuity are highly correlated (~0.76). That's expected (larger loans mean larger payments). We keep both for now, because both contain real business meaning (requested principal vs periodic obligation). If we build linear-style models later, we can regularise or build ratios like loan_annuity / income to express affordability directly.

After removing nr_children, we call the dataset df_step4.

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Data Preparation

Categorical encoding

Now we convert everything into numeric features in a controlled way and finish the dataset. We split features into six groups and encode each group appropriately.

Binary flags - convert to 0/1 integers.

Nominal categoricals - one-hot encode. We keep "Unknown" (from the imputed income_type) as its own dummy column. We don’t hide it or merge it. That way the model can learn that not declaring income type might be a red flag. For linear models, we can drop one dummy per group to avoid perfect collinearity; for tree-based models we can keep all.

Ordinal categoricals - keep them as integers, because the order matters. We do not one-hot them.

Weekday - weekday_of_application treat as nominal (one-hot).

Hour - this is cyclical (23:00 is “close” to 00:00). We encode it as two numeric features hour_sin and hour_cos. This preserves circular structure across midnight.

Engineered numeric features

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Data Preparation

Final ABT

```
=== FINAL MODEL DATA ===
X_ready shape: (7871, 64)
y shape: (7871,)

X_ready preview:
```

	owns_a_car	owns_a_bike	has_an_active_loan	owns_a_house	has_provided_a_home_number	was_reachable_at_work
0	0	0	0	0	0	0
1	1	0	0	1	0	0
2	0	1	1	0	1	1
3	0	0	1	1	0	0
4	0	0	1	1	0	0

5 rows × 64 columns

At this point, we assemble the final version for modelling:

X_ready: Final ABT (Analytical Base Table)

y: The target has_defaulted, extracted as a separate vector.

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Data Preparation

Scaling data

MinMax-scaled feature matrix shape: (7871, 64)

	income_log_mm	credit_log_mm	loan_annuity_log_mm	employment_days_log_mm	residence_density_mm	age_in_days_mm	consecutive_days_of_employment_mm
0	0.630867	0.951296	0.866633	0.487588	0.272869	0.209336	0.021894
1	0.861280	0.500198	0.598508	0.494696	0.243683	0.135923	0.023110
2	0.487034	0.564153	0.623288	0.860161	0.421903	0.305074	0.354619
3	0.861280	0.766968	0.767559	0.861059	0.335355	0.360394	0.356991
4	0.630867	0.910092	0.817853	0.478532	0.630952	0.358539	0.020434

5 rows × 64 columns

Some models are sensitive to raw feature ranges. For example, credit_log and employment_days_log live on different numeric scales even after engineering. Distance-based models (like k-NN) and certain neural-style models tend to behave better if everything is brought to a common [0,1] range.

We handle that with **min-max scaling**: We apply it only to the continuous-style columns. We do NOT scale one-hot dummies or binary 0/1 flags. We build a parallel dataset **X_ready_minmax** - min-max scaled version of the continuous columns (with _mm suffix).

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Target imbalance

Our target has_defaulted is heavily **imbalanced**: about ~7% defaults (1) vs ~93% non-defaults (0). So a dumb model could predict "0" for everyone and still look "accurate." That's not acceptable.

In modelling we will handle that in one or more of these ways:

class weights (penalise missing a default more than missing a non-default),

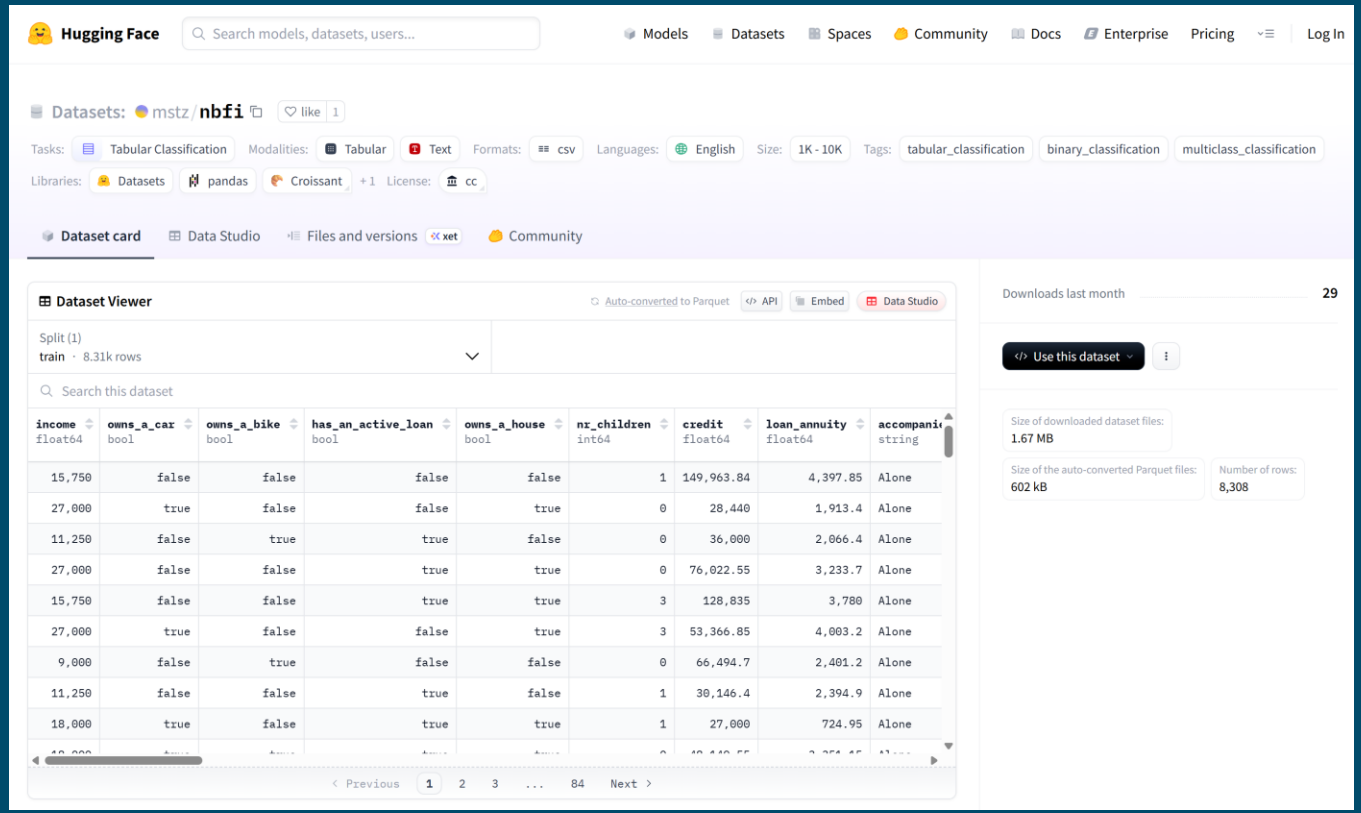
oversampling defaulters (replicate / synthesize more rows with has_defaulted = 1 in the training split),

undersampling non-defaulters (downsample some of the 0s in training),

and using proper metrics like **recall on the default class, precision, ROC AUC, PR AUC** — not just plain accuracy.

We do not try to "balance" the dataset here by deleting good rows. We keep everyone in the ABT and handle imbalance when we train.

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Click on image to open dataset

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